

Project Evaluation: Expense Tracker

Test Guide

This document provides a structured, step-by-step testing guide for the Full-Stack MERN Expense Tracker project. It includes sample data to evaluate the complete user journey, from authentication to data visualization.

Step 1: User Registration (Sign Up)

To evaluate the authentication flow and password hashing, please create a new account using the following test data:

Field	Test Value
Full Name	John
Email	john@gmail.com
Password	12345678

Expected Result: The account is successfully created, a JWT token is generated, and you are redirected to the main Dashboard.

Step 2: Adding Income

Navigate to the **Income** page to test the creation of income records. Use the "Add Income" form with the following two records:

Field	Record 1 (Primary Income)	Record 2 (Secondary Income)
Title / Source	Monthly Salary	Freelance Project
Amount	8000	1500
Date	2026-09-01	2026-09-10
Category	Salary	Development
Icon (Emoji)	💰	💻

Expected Result: Success notifications appear, the income list updates immediately, and the Bar Chart reflects the new earnings.

Step 3: Adding Expenses

Navigate to the **Expense** page and use the "Add Expense" form to insert the following records:

Field	Record 1	Record 2
Category	Groceries	Internet Bill
Amount	600	200
Date	2026-09-12	2026-09-14
Icon (Emoji)	🛒	🌐

Expected Result: Success notifications appear, the expense list updates immediately, and the Line Chart reflects the spending trends over time.

Step 4: Dashboard Verification

Navigate back to the **Dashboard** page. This step verifies that the backend aggregation pipelines (using MongoDB Aggregate) are functioning correctly:

- **Total Income:** Should display the exact sum of the incomes added in Step 2 (e.g., 9500).
- **Total Expenses:** Should display the exact sum of the expenses added in Step 3 (e.g., 800).
- **Total Balance:** Should accurately reflect the mathematical difference (Income - Expenses = 8700).
- **Recent Transactions:** The grid should display a combined, chronologically sorted list of the latest incomes and expenses.

Step 5: Data Export & Deletion

- **Export to Excel:** On either the Income or Expense page, click the Download button. Verify that an .xlsx file is downloaded containing the test records.
- **Delete Record:** Click the trash/delete icon on any specific income or expense card. Verify the confirmation prompt, the removal of the item from the list, and the automatic chart update.